F7AB - Actuarial and Financial Mathematics
Tutorial 3 – Nominal and Real interest rates

Question 1

1. Prove that
$$a_{\overline{n}|}^{(m)} = \frac{1 - v^n}{i^{(m)}}.$$
2. Hence show that
$$s_{\overline{n}|}^{(m)} = \frac{(1+i)^n - 1}{i^{(m)}}.$$

Question 2

1. State in words the meaning of the notation $\ddot{s}_{\overline{10}|}^{(12)}$.
2. State in words the meaning of the notation $600\ddot{a}_{\overline{10}|}^{(12)}$.
3. Prove that
$$\ddot{s}_{\overline{n}|}^{(m)i} = \frac{(1+i)^n - 1}{d^{(m)}}.$$
4. Hence show that
$$\ddot{a}_{\overline{n}|}^{(m)} = \frac{1 - v^n}{d^{(m)}}.$$

Question 3 Evaluate the following.

1. $s_{\overline{10}|}^{(2)\,0.05}$
2. $a_{\overline{17}|}^{(6)\,0.1}$
3. $\ddot{s}_{\overline{25}|}^{(4)\,0.035}$
4. $\ddot{a}_{\overline{7}|}^{(12)\,0.12}$
5. $\ddot{a}_{\overline{\infty}|}^{(4)\,0.06}$

Question 4 [Relationship between real and money yields (I).]

1. An investor pays £100 for an asset. Exactly one year later, the investor sells the asset for £110. Over the same year, the RPI has increased by the rate 3%.
   (a) Calculate the investor's money yield.
   (b) Calculate the investor's real yield.
2. An investor pays £1 for an asset. Exactly one year later, the investor sells the asset for $£(1+i)$, with $i > -1$. Over the same year, the RPI has increased at the rate q.
   Show that the real yield i' on the transaction satisfies
$$\boxed{1 + i' = \frac{1+i}{1+q}}$$

Question 5 [Relationship between real and money yields (II).] Assume that time is measured in years. At time 0, an investor pays £100 for an investment which produces a cashflow of £50 at time 1 and a cashflow of £60 at time 2. There are no further cashflows or value from the investment.

1. Calculate the investor's annualised money yield i.
2. Calculate the investor's annualised real yield i', based on the Inflation Index below.
3. Calculate the average annualised rate of change $\bar{q}$ in the Inflation Index.

Time	0	1	2
Inflation Index at time	100	104	112

4. Explain why, in general,
$$1 + i' \neq \frac{1+i}{1+\bar{q}}\,.$$

5. How must the inflation rate change as a function of time to have the equality $1+i' = \frac{1+i}{1+\bar{q}}$?

Question 6 A particular transaction will provide an investor with a yield of 6.400% per annum.

1. Assuming that over the period of the transaction the annual rate of inflation will be constant and equal to 2.800%, calculate the investor's real yield on the transaction.

2. Calculate the possible values of the constant annual rate of inflation q such that the investor obtains a real yield on the transaction of at least 3.750% per annum.

Question 7 A loan of £25 000 was issued in February 2010 and was repaid at par after three years. Interest accrues on the loan at the rate of 8% per annum and is payable annually in arrears.

The value of an Inflation Index at various times was as follows.

Date	Feb 2010	Feb 2011	Feb 2012	Feb 2013
Inflation index at date	205.0	215.6	223.5	231.5

Calculate the real rate of return on the loan by interpolating between two suitable interest rates which have a difference of at most 0.5%.

Answers: Q3.1. 12.73320; Q3.2. 8.34959; Q3.3. 39.79822; Q3.4. 4.85527; Q3.5. 17.28711: Q4.1(a). 10%; Q4.1(b). 6.796%; Q5.1. 6.394%; Q5.2. 1.077%; Q5.3. 5.830%; Q6.1. 3.502%; Q6.2. $q \leq 2.554\%$; Q7. 3.670%.